

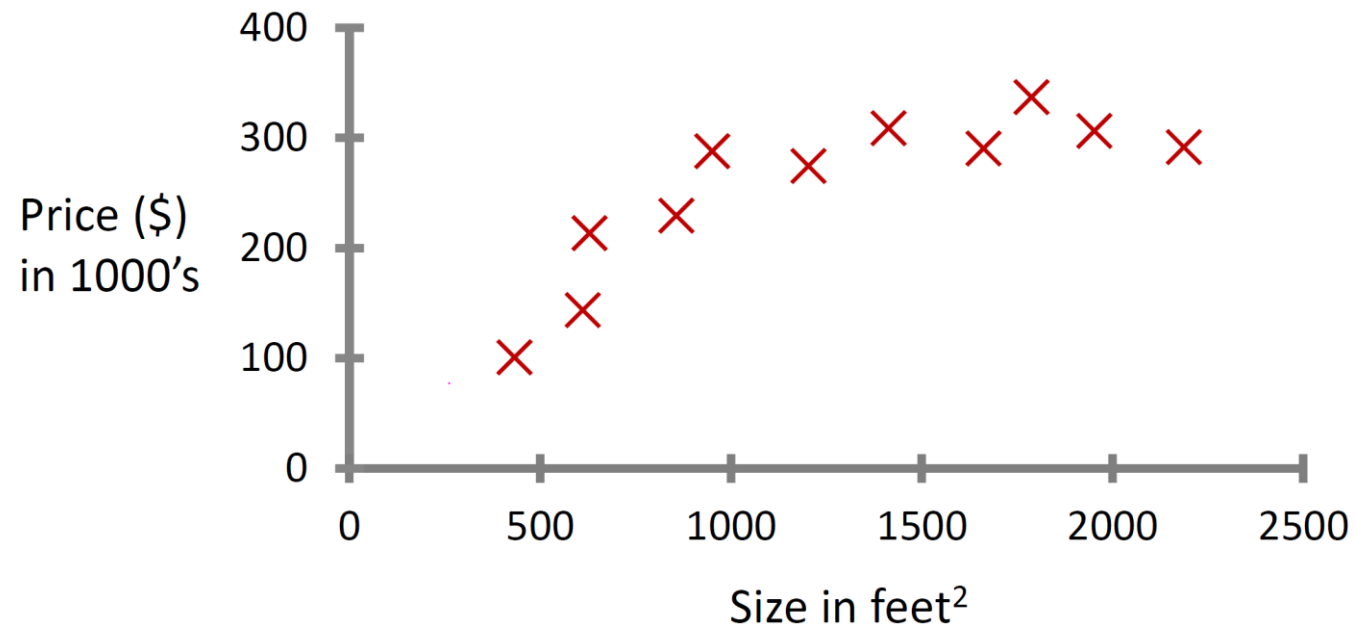
Logistic Regression

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Housing price prediction

Suppose you have a friend with a house of 750 feet². How can a learning algorithm help you?



Classification

The classification problem is like the regression problem, except that the values we want to predict take on only a small number of discrete values.

Some examples:

Spam/ Not Spam

Image with a face / image without a face

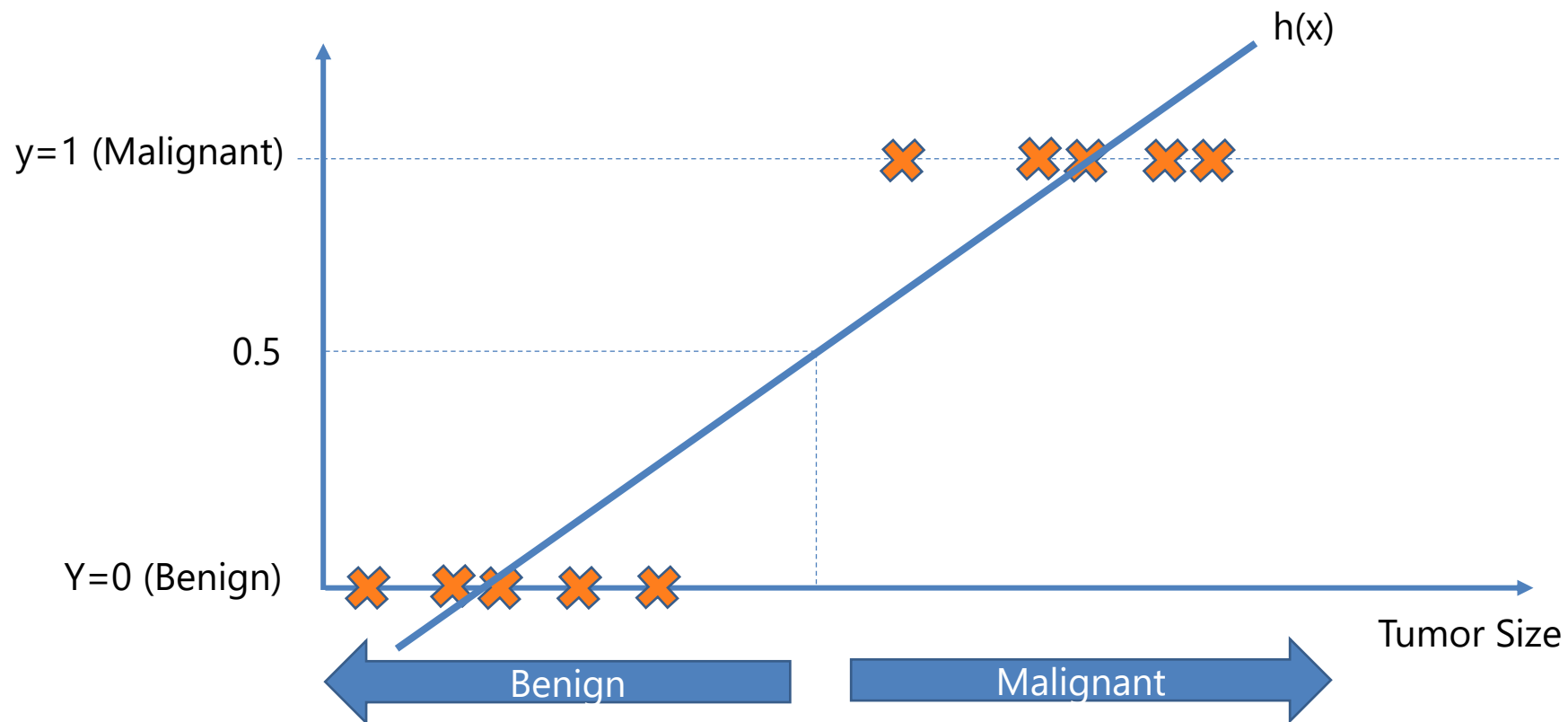
Malignant tumor / Benign tumor

Binary classification problem: predictions can take on only two values, 0 (usually called "Negative Class") or 1 (usually called "Positive Class").

Classification with linear regression

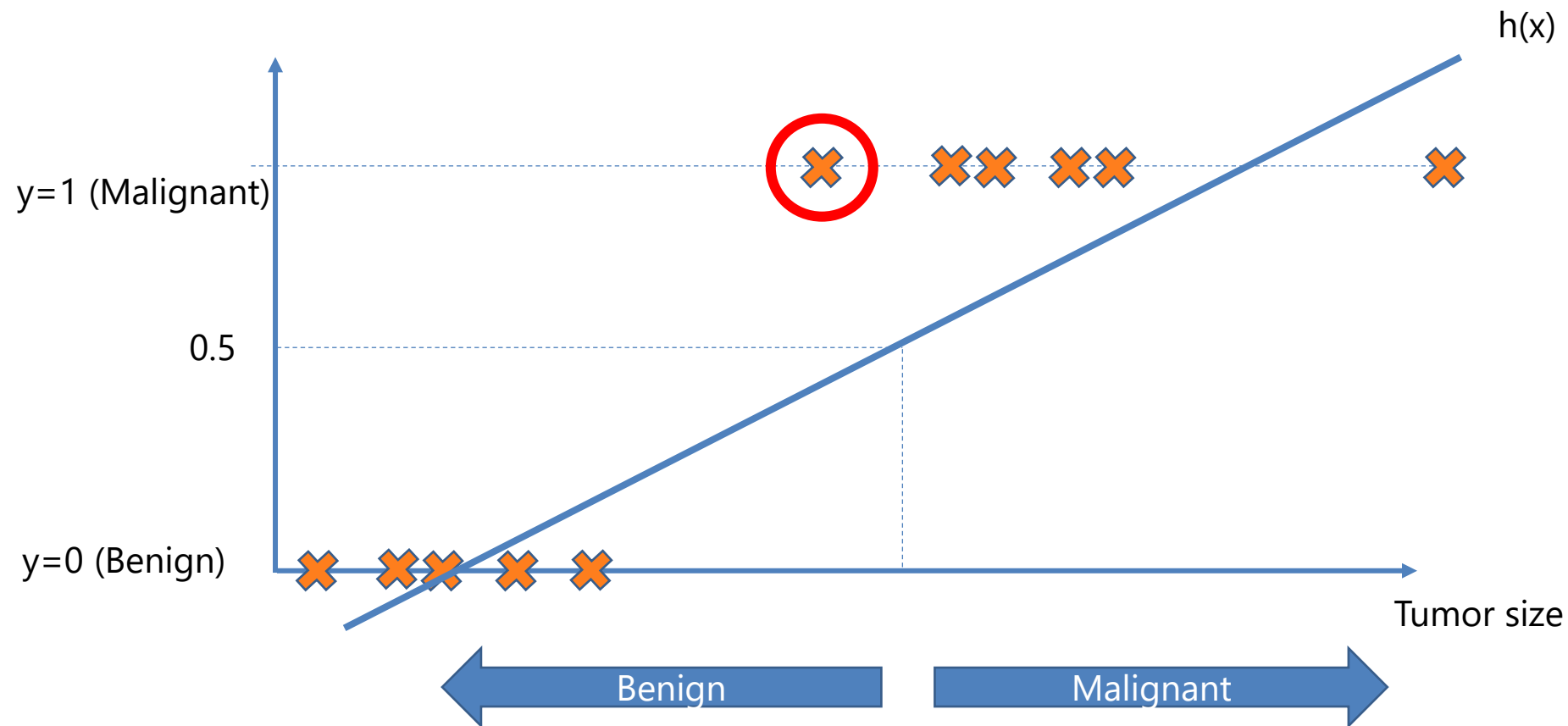
To attempt classification, one method is to use linear regression and map all predictions greater than 0.5 as a 1 and all less than 0.5 as a 0.

Tumor classification based on the tumor size



Classification with linear regression

However, this method doesn't work well because classification is not actually a linear function



Logistic Regression

Applying linear regression to a classification problem is often not a great idea.

In fact, if you use a linear regression, predictions or hypothesis $h(x)$ can take values larger than 1 or smaller than 0 ($h(x) > 1$ o $h(x) < 0$)

Solution: Logistic Regression, which the predictions are always between zero and one.

Logistic Regression, even the regression term appears in this name, is a classification algorithm.

Logistic Regression

Let's change the form for our hypotheses $h_{\theta}(x)$ to satisfy $0 \leq h_{\theta}(x) \leq 1$

We use "Sigmoid Function", also called the "Logistic Function"

$$g(z) = \frac{1}{1 + e^{-z}}$$

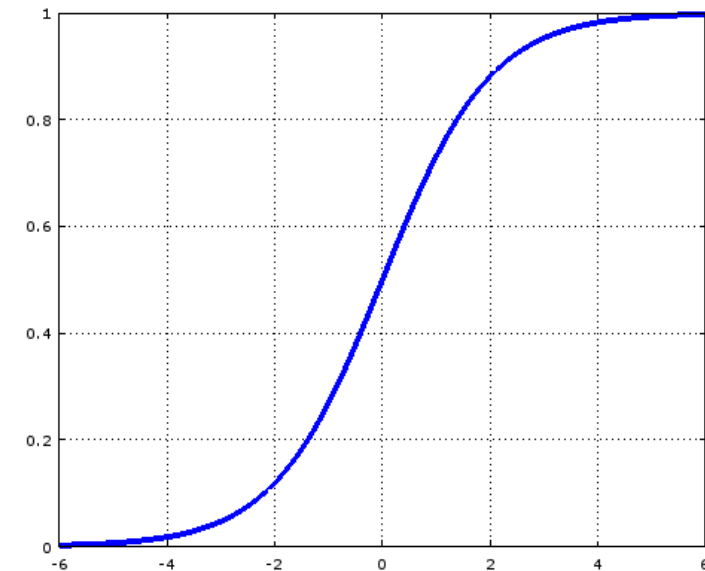
The function $g(z)$ maps any real number to the (0,1) interval.

Linear Regression:

$$h_{\theta}(x) = \theta^T x$$

Logistic Regression:

$$h_{\theta}(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}}$$



Logistic Regression

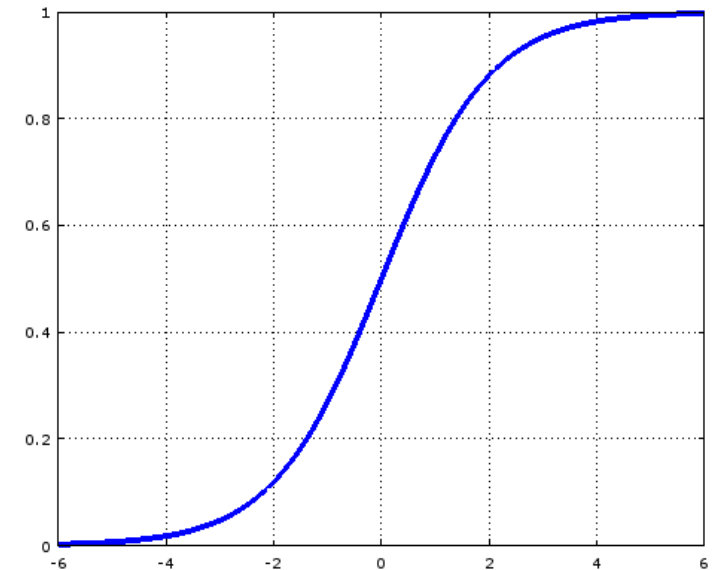
To get our discrete 0 or 1 classification, we map the output of the hypothesis function as follows:

$$h_{\theta}(x) \geq 0.5 \rightarrow "1"$$

$$h_{\theta}(x) < 0.5 \rightarrow "0"$$

The function $g(z) > 0.5$ when $z > 0$, then

$$h_{\theta}(x) = g(\theta^T x) > 0.5 \text{ when } \theta^T x > 0$$



The decision boundary, in a linear case, is a line that separates the area where prediction is equal to 0 and where prediction is equal to 1

Cost Function

Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)}), \}$

The goal is to find θ so
$$\begin{cases} h_{\theta}(x) \geq 0.5 & \text{if } y = 1 \\ h_{\theta}(x) < 0.5 & \text{if } y = 0 \end{cases}$$

We can define the following cost function:

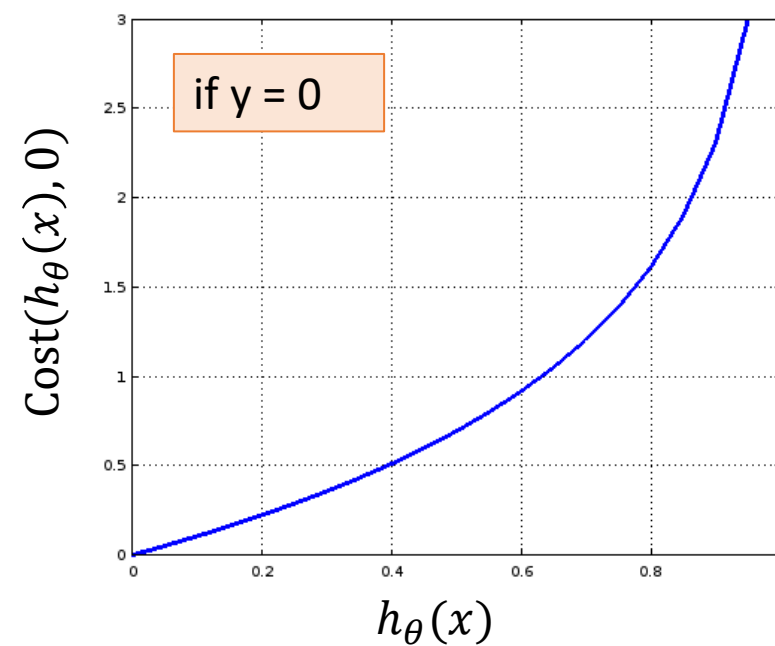
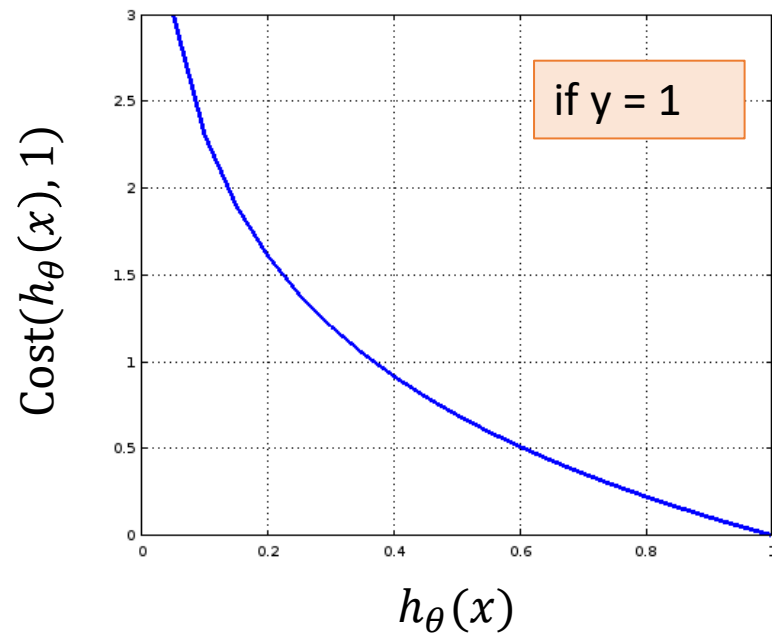
$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x), y)$$

where $\text{Cost}(h_{\theta}(x), y) = -y \log(h_{\theta}(x)) - (1 - y) \log(1 - h_{\theta}(x))$

This cost function is convex and is derivable respect to θ , so we can use Gradient descent to solve the minimization problem.

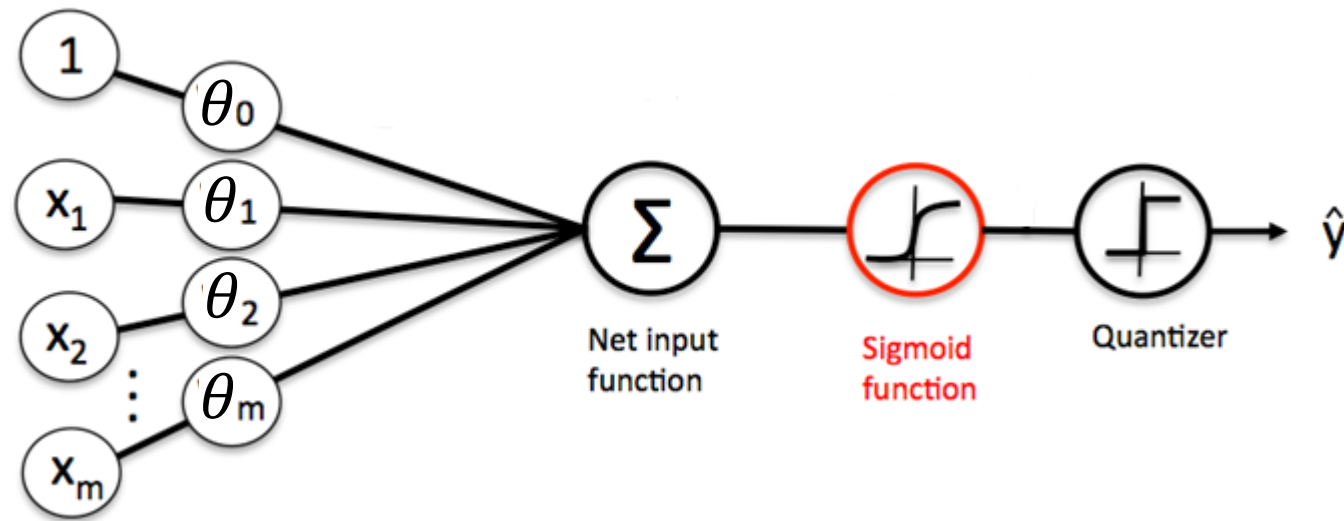
Cost Function

$$\text{Cost}(h_{\theta}(x), y) = -y \log(h_{\theta}(x)) - (1 - y) \log(1 - h_{\theta}(x))$$



Case $y=1$, the cost function will be 0 if our hypothesis function $h_{\theta}(x)$ outputs 1. if our hypothesis approaches 0, then the cost function will approach infinity.

Logistic Regression



Logistic Regression: Decision Boundary

Let's consider a 2D case $h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$

with $\theta = [\theta_0, \theta_1, \theta_2]^T, x = [1, x_1, x_2]^T$

Using the Logistic Regression Algorithm, we can obtain

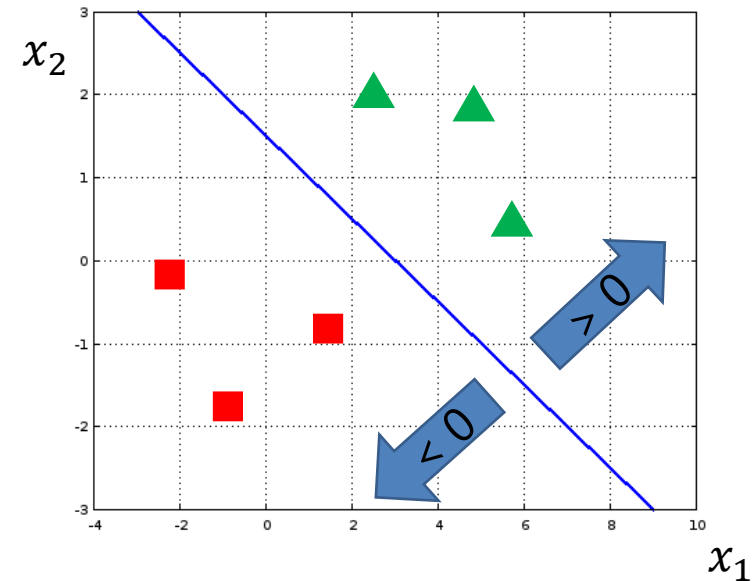
$$\theta = [-3, 1, 2]$$

Then the decision boundary is

$$h_{\theta}(x) = 0.5 \Rightarrow \theta^T x = 0 \Rightarrow -3 + x_1 + 2x_2 = 0$$

Remember:

- $h_{\theta}(x) = g(\theta^T x) > 0.5$ when $\theta^T x > 0$



Decision boundary ($\theta^T x = 0$)

Logistic Regression with no-linear decision boundaries

The hypotheses function does not need to be linear, it could be a no linear function. If so, decision boundary could not be a line but to be another function.

Suppose

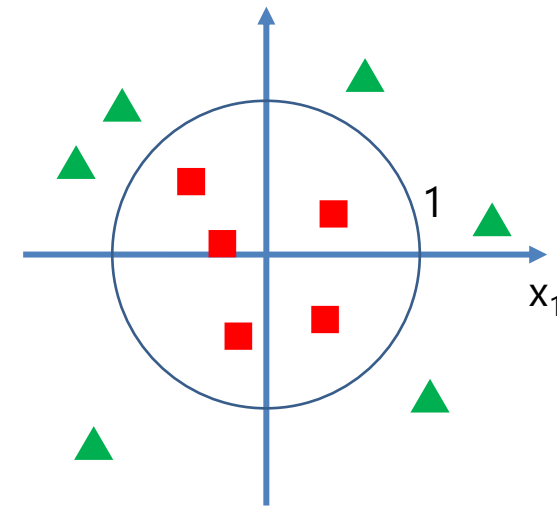
$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2)$$

Using the Logistic Regression, we can obtain:

$$\theta = [-1 \ 0 \ 0 \ 1 \ 1]$$

Then the decision boundary is

$$h_{\theta}(x) = 0.5 \Rightarrow \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2 = 0 \Rightarrow -1 + x_1^2 + x_2^2 = 0$$



Decision boundary

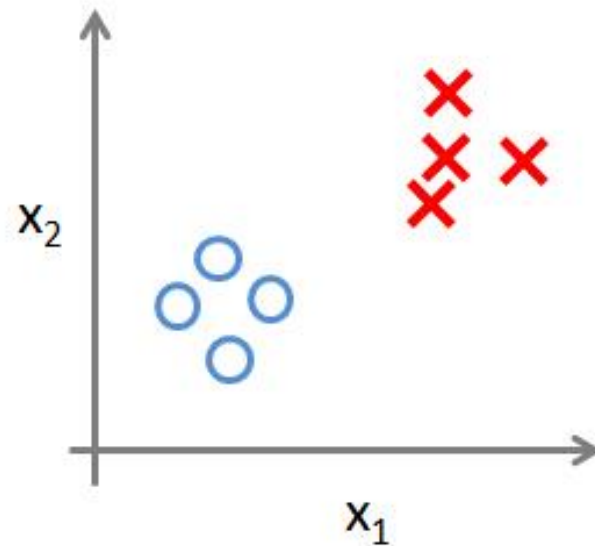
Multiclass Classification

Multiclass classification: classification problem when we have more than two categories

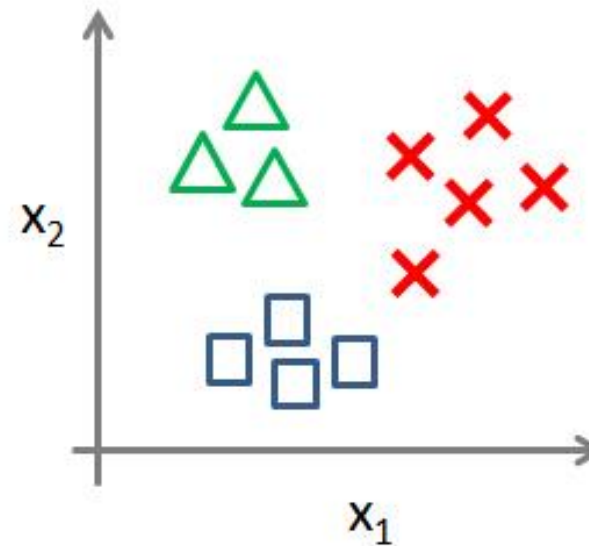
DeepNin?

For example : Weather: Sunny, Cloudy, Rain

Binary classification:



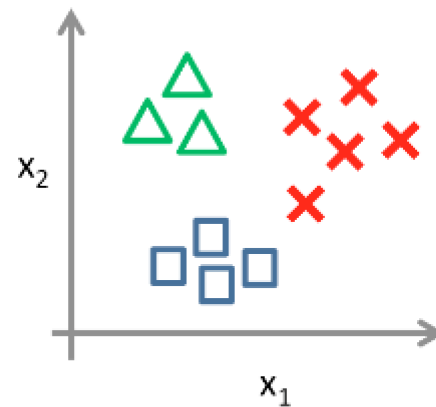
Multi-class classification:



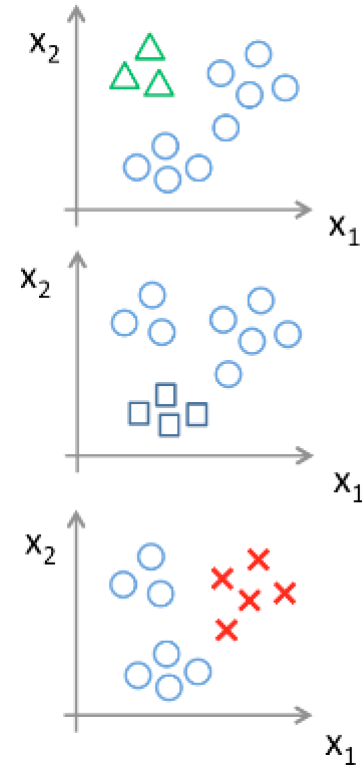
One-vs-all Approach

Suppose we have 3 categories

One-vs-all (one-vs-rest):



Class 1: 
Class 2: 
Class 3: 



To make a prediction on a new x , pick the classifier that maximizes $h_{\theta}(x)$